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CSE

Class Test - CO4 AT1

① parsing Table conflict in LL

Construct three address code for the following

(a c b) and (c c d) or (a > d) then

Z = X + Y * Z

else

Z = Z + 1

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Step 1 :-

Condition is :-

a c b
c c d
a > b

the structure is (a < b) and (c c b) (or) (a > b)

Step 2 :- Create table L1, L2, L3, L4, L5

Step 3 : Generate TAC for the condition

L1 : if a < b goto L2
goto L5

L2 : if ~~a~~ c < b goto L4
goto L3

L3 : if a > b goto L4 . goto L5

Step 4 :- Generate TAC for the then part

① $z = x + y * z$

L_4 :-

$t_1 = y * z$

$t_2 = x + t_1$ goto L_6

ii] $z = z + 1$

L_5 :- $t_1 = z + 1$

$z t_2 = t_1$

final :-

L_1 = if $a < b$ goto L_2

goto L_5

L_2 : if $a < b$ goto L_4

goto L_3

L_3 : if $a > b$ goto L_4

goto L_5

L_4 : $t_1 = y * z$

$t_2 = x + t_1$

$z = t_2$

goto

L_5 : $t_1 = z + 1$

$z = t_1$

Step 5 : final backpatched code

if a<b goto 3
goto 5
if c<d goto 7
goto 5
if e<f goto 7
goto 8.
True . false.

3) Construct three address code using the following statment (or) production for boolean expression (a>b) or (c<d) or (a<f)

$B \rightarrow B_1 \| B_2$

$B \rightarrow B_1 \; \& \& \; B_2$

$B \rightarrow E_1$ Setop E_2

$B \rightarrow$ true

$B \rightarrow$ false.

A) Step 1 :-

a>b
c<b
e<f

Assume expression is : $(a>b) \| c<d \| e<f$

(2) Translate (a<b) and (c<b) or (e<f) into three adress statment using back poching.

Step 1 :- Condition is :

$a<b$
$c<d$
$e<f$

The structure is (a<b) and (c<d) or (e<f)

Step 2 :-

if a<b goto -
goto -
if c<d goto -
goto -
if e<f goto
goto -

Step 3 :- Handle AND

for (a<b) AND (c<d) if a<b is true.

if a<b goto 3.

Step 4 : Handle OR

if the first AND expression is false. The Second OR operand e<f must be evalulated

three

Step 2 : Generate three address code for a > b

if a>b goto true
goto L_1

Step 3 : c<d,

if c<d goto L_1, true
goto L_3

Step 4 : e<f,

if e<f goto L.true
goto false.

final three - Adress code

if a>b goto L.true
goto L_2
if c>c&d goto L.true
goto L_3
if e<f goto L.true

L.true : // Boolean expression is true
goto L.END

L.false : // Boolean expression is false.